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Unit 1 - Understanding Networks and the Internet

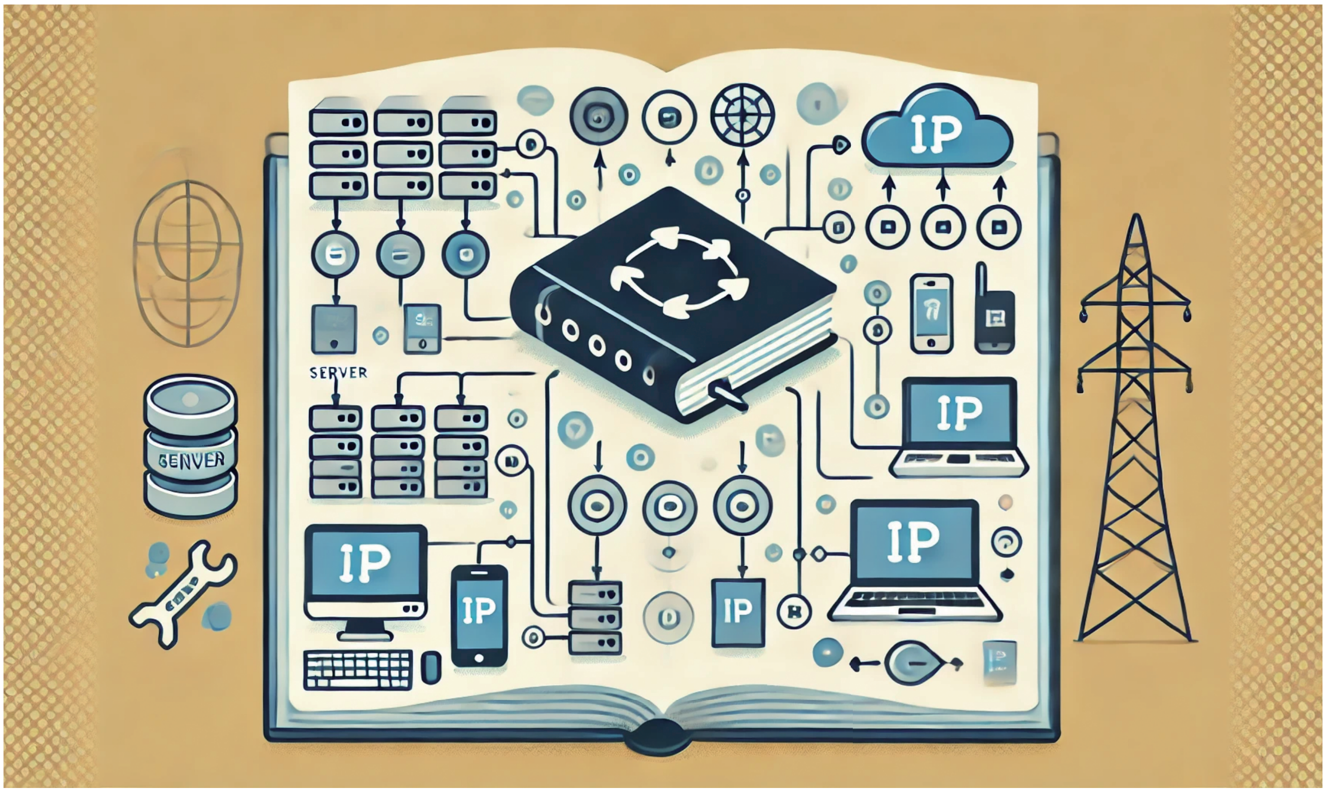
PDF Documents

PDF Document 1: 1738211794881-Chapter 1 - Unit 1 - Understanding Networks and the Internet.pdf

This PDF document is included in the unit materials.

UNIT 1 UNDERSTANDING NETWORKS AND THE INTERNET

What is a Network?



Have you ever wondered how computers communicate and share information? The answer lies in **networks**! A network is a group of computers or devices linked to exchange information and resources. Networks can be small, like the ones in your home or school, or huge, connecting millions of devices worldwide.

When you send an email, play an online game, or search for a video, your device sends and receives information through a network. Networks allow you to connect with people, access data, and share information quickly and efficiently.

HOW DOES THE INTERNET WORK?

The **Internet** is the world's largest network, connecting millions of computers, smartphones, and other devices. It allows people to share information, communicate in realtime, and access websites and services anywhere. But how does this vast network work?

The internet sends data through wires, cables, satellites, and wireless connections, like **WiFi**. When you use your computer or phone to visit a website, your device sends data packets through this system to the server where the website is stored. The server then sends data packets back to your device, and your screen displays the website or content you requested.

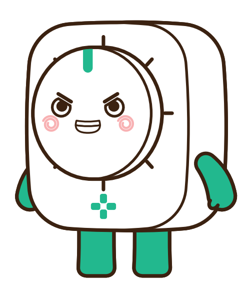
KEY PARTS OF THE INTERNET

Understanding how the internet works involves knowing about some important parts that help it run smoothly:

1 **Servers:** Servers are powerful computers that store websites, applications, and data. When you type a web address into your browser or click on a link, your device sends a request to a server. The server processes this request and sends the information you need back to your device in tiny pieces called packets.

2 **Routers:** Routers help direct data between different networks. They act like traffic controllers, ensuring that your message or search request takes the fastest and most efficient route to its destination. Routers ensure that data gets where it needs to go, even if many networks and devices are involved.

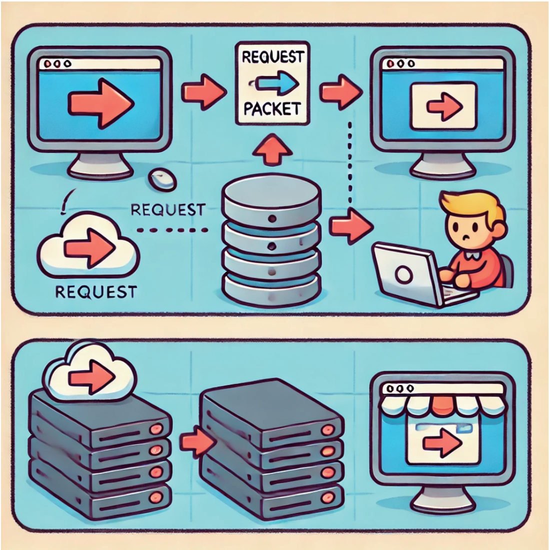
3 **IP Addresses:** Every device connected to the Internet has a unique IP address, which stands for Internet Protocol address. Think of it as your device's home address on the Internet. Just as the mail carrier needs to know where to deliver your letters, data packets need an IP address to know where to go and how to return to you.



How Data Travels

When you use the internet, data moves in tiny pieces called packets. These packets carry information from your device to the server and back.

For example, when you click on a website link:



1. Your computer sends a request packet to the server.
2. The server receives the packet, processes the request, and returns response packets to your computer.
3. Your computer puts these packets together to display the website.

WHY NETWORKS AND THE INTERNET ARE IMPORTANT

Networks allow us to connect with people, access information, and share data quickly. The Internet has become a part of everyday life and is used in schools, homes, businesses, and even cars. Understanding how networks work helps you use technology better and stay informed.

How Data Travels

Create a simple diagram showing how devices connect in a network. Include:

Computers and smartphones.

A router connecting the devices.

A server labeled as the source of a website.